Engineering Closure: A First-Order Mass, Power, Braking, and Reproduction Model for a Slow Self-Replicating Interstellar Probe

S. Stone

Working draft — analytical first pass

Contents

Working draft — analytical first pass. Fourth in a series. The companion papers treat the vehicle ("Growth, Not Speed"), the payload ("The Payload"), and the seed-to-factory bootstrapping problem ("From Seed to Factory"). Those papers repeatedly defer four quantitative budgets to "the engineering follow-on"; this paper is that follow-on, at first order. It supplies model structure and order-of-magnitude estimates, not computed results across the real stellar catalogue — the code-backed computation is carried out in the companion computational paper ("Engineering Closure, Computed").

Abstract

The conceptual papers in this series argue that a probe optimized for persistence and growth rather than speed should be slow, self-repairing, and self-replicating, and they defer the quantitative work to a follow-on. This paper supplies the first-order version of that follow-on, organized around four interlocking budgets. The **mass-and-power budget** decomposes a notional seed into subsystems and finds that nuclear fuel is negligible while shielding, radiators, the magnetic-sail braking structure, and — dominating everything — the in-situ mining and manufacturing plant set the scale, placing a plausible seed between ~10^3 kg (a minimal bootstrap package) and ~10^7 kg (a full Freitas-class factory). The **braking budget** gives the load-bearing deceleration claim its first quantitative check: a magnetic sail braking against the interstellar medium

stops in a distance $d = m/(2\rho A)$ that is *independent of cruise speed*, so a ~100 km effective-radius sail halts a tonne-scale seed in years over ~0.005 ly, while smaller sails take centuries over fractions of a light-year — feasible, but coupling braking directly to the seed's mass-to-sail-area ratio. The **reproduction budget** casts galactic expansion as a branching process (Harris 1963): expansion requires an effective reproduction number $R\_eff = \Sigma_i p_i V_i$ greater than one, where p_i is per-leg survival and V_i is target viability, and the result is a knife-edge — a few percent of per-leg failure separates exponential spread from extinction. The **closure budget** shows that accessible material is never the binding constraint (a single asteroid belt outmasses any child by 14 to 18 orders of magnitude); the binding constraint is *manufacturing capability*, expressed as a closure ratio whose shortfall is a small "vitamin" fraction of un-makeable parts. We then show the four budgets interlock — a heavier seed needs a larger sail and more material and energy to reproduce, lengthening replication time and, wherever a node's production lifetime is finite, thinning the children it can attempt — R_eff must still clear one — and argue that a self-consistent regime exists at first order (tonne-to-kilotonne seed, ~100 km sail, a kilowatt-class carried reactor growing to megawatt-class manufacturing power in situ, partial closure with a small vitamin set, modest R_eff > 1). Mapping the boundaries of that regime across the real catalogue is the computational work this paper sets up but does not perform.

---

1. Introduction

The first three papers of this series make a connected architectural argument — slow travel for navigability and survivability, nuclear power and self-repair for centuries of operation, braking and settlement for reproduction, and replication rather than velocity as the engine of galactic spread — and each defers the same quantitative obligations to a follow-on: a subsystem mass and energy budget, a quantitative braking model, a resource-and-demographics model of reproduction, and a closure model coupling manufacturing capability to whether a child can be built at all. This paper discharges those obligations at first order.

"First order" is meant precisely. We give the *structure* of each model — the governing relation and the variables it depends on — and order-of-magnitude estimates from it, using numbers already established in the companion papers where they exist. We do not compute distributions across the real 127-star catalogue (the systems within 100 light-years), optimize designs, or close the multi-parameter trade space; that requires extending the project's existing simulator with failure rates, target viability, and closure, and is the natural subject of a subsequent, computational paper. The value of the present first pass is to show that the four budgets are individually tractable and jointly self-consistent, and to identify which terms dominate and which are knife-edged — the information needed to scope the computation that follows. The seed-to-factory *capability* question — how a small package bootstraps a factory — and the biological-capability question are treated separately in the bootstrapping paper and are not duplicated here; this paper takes manufacturing capability as a parameter (the closure ratio) and asks what budgets it must satisfy.

2. The Mass-and-Power Budget

We reason about a *notional seed*: the object launched toward a target, which on arrival brakes, settles, and grows into the factory that builds the next seeds. Its mass is the central design variable because it sets the braking budget (Section 3) and the material and energy required to reproduce it (Sections 4–5).

Decomposed by subsystem, the budget separates cleanly into a negligible part, a moderate part, and a dominant part. The **negligible** part is nuclear fuel: as the vehicle paper established, ~16.5 kg of ^{235}U supplies 4 kW electric for 300 years (at ~28% conversion and ~10% burn-up), so fuel mass never drives the design. The **moderate** part is the power-and-thermal plant: a kilowatt-class space-fission reactor of the KRUSTY type (Poston, Gibson et al. 2020) masses on the order of hundreds of kilograms to a tonne including shielding and Stirling or Brayton conversion — of order 10^2 kg kW^{-1} at kilowatt scale once shielding is included, improving substantially at higher power; its radiators follow the blackbody floor $A \approx P/(\varepsilon\sigma T^4)$, about 11 m² to reject the reactor's full ~14 kW rated thermal output at a 400 K radiator (the reactor's rated full-power output, used in manufacturing; actual cruise draw is 50–150 W, requiring well under 0.5 m² in steady state, though the 11 m² is carried because it is sized for full-power pulse events), and rising to hundreds of m² at megawatt manufacturing power. Computation and the cognitive payload are, by comparison, light — of order 10^2 kg of radiation-hardened electronics and storage. The **dominant** part is everything that distinguishes a settlement from a spacecraft: the magnetic-sail braking structure (Section 3), and above all the *in-situ mining and manufacturing plant* — excavation, beneficiation, refining, fabrication, and the launch infrastructure for the next generation. It is the manufacturing plant, not the reactor or the payload, that sets the seed's true mass and the difficulty of reproducing it.

This decomposition is why the plausible seed mass spans four orders of magnitude. A *minimal bootstrap package* — the mobile seed defined in the vehicle paper, carrying only enough to brake, survive, and begin building from local material — might be ~10^3–10^4 kg, deferring most manufacturing capability to what it constructs after arrival. A *full self-contained factory* in the spirit of von Neumann's universal constructor (von Neumann & Burks 1966), the NASA 1980 self-replicating lunar study (NASA 1982), and Freitas's interstellar adaptation (Freitas 1980) is ~10^6–10^7 kg. The architecture strongly prefers the low end, because seed mass propagates into every other budget; the bootstrapping paper's staged-closure cascade is precisely the strategy for keeping the launched seed small while the *arrived* system grows large. We therefore carry a reference seed of ~10^3–10^4 kg through the budgets below (the computational companion's reference point lands at ~3,700 kg), noting where a heavier seed changes the conclusions.

3. The Braking Budget

Braking is the architecture's load-bearing claim — without it there is no settlement and no reproduction — and the vehicle paper originally gave it only a placeholder, t_brake ~ mv/F_drag (its current revision imports the model closed here). Here we close that relation at first order for the most plausible mechanism, a magnetic sail decelerating against the diffuse plasma it sweeps up (Andrews & Zubrin 1990; Freeland

2015; see also Crawford 1990 on interstellar deceleration).

Treating the sail as a drag device of effective area A in a medium of mass density ρ, the swept-momentum drag force is $F \approx \rho v^2 A$, giving a deceleration $a = F/m$, a braking time $t_{brake} = m/(\rho v A)$, and — the instructive quantity — a braking distance

$$d = m / (2\rho A),$$

which is *independent of the cruise speed v*. Formally d is the characteristic (momentum-halving) arrest distance — under the drag law the residual velocity decays exponentially, and the terminal crawl is absorbed in stellar-wind capture. The stopping distance depends only on the seed's mass-to-sail-area ratio and the medium density, a "ballistic coefficient" for interstellar braking. Using the local interstellar density $\rho \approx 3.3 \times 10^{-22}$ kg m^{-3} ($n \approx 0.2$ cm^{-3}; Redfield & Linsky 2008; Frisch et al. 2011) and a 450 km s^{-1} cruise, the numbers are concrete: a tonne-scale seed (10^3 kg) with a 100 km effective-radius magnetic sail stops in ~7 years over ~0.005 ly; the same seed with a 30 km sail takes ~74 years over ~0.056 ly, and with a 10 km sail ~670 years over ~0.5 ly. A heavier 10^4 kg seed scales the distance and time up by ten, so a 100 km sail still stops it in ~67 years over ~0.05 ly, while a 10 km sail would require ~5 ly and millennia.

Three conclusions follow. First, braking is *feasible* for plausible magnetic-sail radii of tens to hundreds of kilometres, which the magsail literature supports (Freeland 2015; Gros 2017), with a deceleration phase of years to centuries spread over a fraction of a light-year — a substantial but not prohibitive part of a millennial mission, and one that reinforces rather than threatens the slow-and-patient thesis. Second, because $d \approx m/(2\rho A)$, the design lever is the sail-area-to-mass ratio, not the speed: a small, gossamer-sailed seed brakes easily, a heavy compact one does not, which is a further argument for the minimal bootstrap package of Section 2. Third, the diffuse-ISM estimate is conservative for the final approach: within the target's astrosphere the stellar wind is orders of magnitude denser (though only over a short path), and gravitational capture and a modest onboard nuclear-electric burn handle the terminal arrest — where the magsail's thrust is weakest, since sail performance degrades at low relative velocity (Perakis & Hein 2016) — so the ISM magsail does the bulk Δv and the stellar environment the final capture. A quantitative deceleration model across real stellar-wind environments remains future work, but the first-order budget shows the claim survives contact with the numbers.

4. The Reproduction Budget

Galactic expansion is correctly modelled not by replication time alone but as a *branching process* (Harris 1963), the same Galton-Watson structure that governs extinction of family names and epidemic spread; a percolation-threshold version of the same idea was applied to interstellar colonization by Landis (1998). The controlling quantity is the effective reproduction number,

$$R_{eff} = \Sigma_i \, p_i V_i,$$

the expected number of *successful* child settlements produced by one settled node, summed over candidate targets i, where p_i is the probability that a child dispatched to target i survives the whole chain — manufacture, launch, cruise, braking, and settlement — and V_i is the viability of the target itself: its accessible small-body mass, volatile inventory, metallicity, fissile or fertile availability, and braking environment. The branching-process result is sharp: if R_eff ≤ 1 the lineage goes extinct with probability one, regardless of how short replication time is; if R_eff > 1 extinction is no longer certain — the lineage survives with positive probability, and surviving lineages expand exponentially — and only then do replication time, travel time, and target density set the *rate*. The resulting galaxy-filling times are of order 10^7–10^8 yr — slower than Hart's (1975) ~10^6 yr for a fast 0.1c fleet, comparable to Tipler's (1980) ~3×10^8 yr.

This reframes the demographic claim of the conceptual papers and exposes its sensitivity. Per-leg survival multiplies — $p_i \approx$ p_manufacture × p_launch × p_cruise × p_brake × p_settle — so a few percent failure at each of five stages compounds to a substantial total, and with each node typically able to reach only a handful of viable neighbours, R_eff sits near the knife-edge of one. A design that reduces any single per-leg failure probability, or that raises the number of reachable viable targets, can be the difference between a galaxy-filling lineage and a stillborn one. The first-order conclusion is therefore not a number but a target: the engineering must drive R_eff comfortably above one, and the dominant risks are exactly the failure terms the vehicle paper enumerated (launch, cruise, braking, settlement, replication, and kernel-integrity failures). Computing R_eff and its variance across the real catalogue, with realistic p_i and V_i, is the central deliverable of the computational sequel.

5. The Closure Budget

Whether a child can be built at all is the closure question, and the first-order analysis overturns the intuitive worry. The intuitive worry is *materials*: that a target might lack the matter to build a child. It almost never does. A single asteroid belt masses ~10^{21} kg; a child of 10^3–10^7 kg is smaller by fourteen to eighteen orders of magnitude, and even rare elements present at parts-per-million are available in tonnes. Materials closure is essentially never the binding constraint.

The binding constraint is *capability*, captured by a closure ratio C = (mass the system can manufacture in situ) / (child dry mass). Full closure, C = 1, is extraordinarily hard; the realistic target is partial closure, C close to but below one, with the shortfall (1 − C) carried as a small inventory of "vitamin" parts — the components the system cannot yet make. As the bootstrapping paper argues — drawing on the closure-engineering formalism of Freitas & Merkle (2004) and terrestrial bootstrapping analyses such as Metzger et al. (2013) — the vitamin set is dominated by microelectronics, enriched fissile material, and precision metrology, and the mission's deep objective is to drive the vitamin fraction toward zero over generations. A near-term design study corroborates the shape of this target: Borgue & Hein (2021) reach ~70% replication by mass with microelectronics carried as unreplicated stock — C ≈ 0.7, with precisely this vitamin set. The energy side of closure is comfortable: the reactor that powers cognition and propulsion also powers mining and manufacturing, and over a replication time of 10^3–10^5 years a kilowatt-to-megawatt plant delivers 10^{13}–10^{18} J, ample for refining and fabricating a child whose own construction energy is far smaller. Closure thus reduces to a single hard question — can the system make, or cheaply carry, the few parts it cannot make? — and that question couples directly back to reproduction: a closure shortfall that the vitamin inventory cannot cover drops p_manufacture toward zero, pulling R_eff below one. Closure failure is, formally, just one of the per-leg failures of Section 4.

6. The Coupled Budget: Why "Closure" Means All Four at Once

The four budgets are not independent, and the engineering result is their simultaneous satisfaction. Seed mass (Section 2) sets the braking budget through d = m/(2ρA) (Section 3) and the reproduction cost through the material and energy needed to copy it (Section 5); the power plant sizes both the radiators and the manufacturing throughput, hence replication time; replication time, target viability, and the compounded per-leg survival set R_eff (Section 4), which must exceed one. A change in any one budget ripples through the others: a heavier, more capable seed brakes harder and costs more to reproduce, lengthening replication time — and, wherever a node's production lifetime is finite, lowering R_eff by thinning the children it can attempt — while a lighter seed brakes easily and reproduces cheaply but carries less capability and a larger vitamin fraction, risking a manufacture failure that also lowers R_eff. "Engineering closure," in the full sense, is the requirement that a single design be mass-sane, brakeable, reproducible under achievable manufacturing capability, *and* demographically expansive, all at once.

The first-order finding is that a self-consistent regime plausibly exists: a tonne-to-kilotonne seed, a magnetic sail of ~100 km effective radius, a kilowatt-class carried reactor growing to megawatt-class manufacturing power in situ, partial closure with a small vitamin set, and an R_eff driven modestly above one by keeping per-leg failures low and viable targets plentiful. Each budget, taken alone, admits such a point. What this paper cannot yet do — and what the computational sequel must — is map the *boundaries* of that regime: the seed-mass ceiling above which braking or reproduction fails, the vitamin fraction above which closure collapses, the per-leg failure budget above which R_eff falls below one, and how those boundaries vary across the real distribution of target systems. The architecture closes at order of magnitude; whether it closes with margin is a computation, not an argument.

7. Discussion and Limitations

This is a first-order paper and inherits first-order limitations. Every budget uses a single representative relation rather than an engineered model: the mass budget omits structure, deployment, and integration masses and treats specific masses as constants; the braking model uses a swept-momentum drag law and a single ISM density, neglecting the velocity-dependent and geometry-dependent physics of real magnetic-sail/plasma interaction and the dense-but-brief stellar-wind contribution; the reproduction model treats p_i and V_i as parameters rather than deriving them; and the closure model reduces a vast manufacturing dependency graph to a single ratio. The dominant unknowns are the same three the whole series keeps returning to — autonomous high-closure manufacturing capability, the real magnetic-sail braking environment, and the per-leg failure rates — and all three are precisely the quantities a computational model must supply with distributions rather than point estimates. The knife-edge nature of R_eff means the demographic conclusion is the most sensitive of the four and the most in need of that computation.

Two further limitations are definitional rather than numerical, and both concern what the model's variables are being asked to mean. First, the per-leg survival probability $p_i \approx$ p_manufacture × p_launch × p_cruise × p_brake × p_settle (Section 4) folds kernel-integrity failure implicitly into the same multiplicative chain — inside p_manufacture and p_settle — alongside ordinary physical loss, the same undifferentiated bucket that holds a cracked radiator or a failed launch. But a probe that fails to survive and a probe that survives with a corrupted core are not the same outcome under any measure that matters: the first is a clean subtraction from the population, the second is a live, propagating node with no guarantee that its behavior still conforms to the immutable core. The model has no variable distinguishing "doesn't survive" from "survives without integrity," and collapsing them into a single p_i implicitly prices a corrupted success as no worse than a clean failure — arguably backwards, since the corrupted case is the one the rest of the lineage must detect and contain rather than simply mourn. The ethics paper in this series takes up the normative stakes of this gap; the modeling point here is narrower: R_eff as defined in this paper cannot see the difference, and a computation that wants to bound risk rather than just count survivors will need p_i decomposed into at least a survives/fails/survives-corrupted trichotomy.

Second, the branching-process criterion of Section 4 is binary by construction — extinction below one, expansion above it — and this paper treats that threshold as the whole demographic finding, indifferent to how far above one a design sits. R_eff = 1.01 and the R_eff = 1.85 four-offspring case computed in the computational engineering paper are engineering-equivalent by this criterion: both are "expansion achieved." But they imply very different colonization postures. Near-unity growth leaves centuries to millennia between branching events at a given node — time to observe a target system, decide whether and how to proceed, and let any contacted intelligence respond before the next commitment is made; high R_eff instead produces many simultaneous fronts expanding in parallel, compressing the deliberation time available before the next irreversible step at each of them. Nothing in this paper's framing treats replication rate itself as a design lever that trades expansion speed against the restraint time preserved at each front; R_eff is optimized as a scalar to be pushed above one, not shaped as a rate with consequences for how much time is available to act before the next branching. That trade is not addressed anywhere in this model, and sits alongside the viability-term and vitamin-fraction gaps the series' ethics paper develops.

None of this undermines the paper's purpose, which is to show that the conceptual architecture is quantitatively *closable* at first order and to identify what the closing computation must contain. The budgets are individually tractable, jointly coupled, and collectively satisfiable in a plausible regime; the remaining work is to replace "plausible" with "bounded."

8. Conclusion

The three conceptual papers argued that a self-replicating interstellar probe optimized for growth over deep time should be slow, nuclear-powered, self-repairing, braking, settling, and reproducing. This paper checks, at first order, that those choices close as an engineering system. Nuclear fuel is negligible and accessible material is effectively infinite, so neither binds; the design is instead bounded by manufacturing capability, by the mass-to-sail-area ratio that governs braking over a speed-independent stopping distance, and by the branching-process requirement that the effective reproduction number exceed one. A self-consistent regime exists at order of magnitude — a light seed, a large sail, a modest reactor, partial closure, and a reproduction number nudged above unity — and the four budgets interlock in a way that makes “closure” a statement about all of them simultaneously. The next step is no longer conceptual: it is to compute these budgets across the real stellar neighbourhood, with realistic failure and viability distributions, and to map the boundaries of the regime in which the architecture closes with margin. That computation, building on the simulator already developed for this project, is carried out in the companion computational paper, “Engineering Closure, Computed” — the work this first-order model is written to enable.

References

Andrews, D. G., & Zubrin, R. M. (1990). Magnetic sails and interstellar travel. *Journal of the British Interplanetary Society*, 43, 265–272.

Borgue, O., & Hein, A. M. (2021). Near-term self-replicating probes — A concept design. *Acta Astronautica*, 187, 546–556.

Crawford, I. A. (1990). Interstellar travel: A review for astronomers. *Quarterly Journal of the Royal Astronomical Society*, 31, 377–400.

Freeland, R. M. (2015). Mathematics of magsails. *Journal of the British Interplanetary Society*, 68, 306–323.

Freitas, R. A. (1980). A self-reproducing interstellar probe. *Journal of the British Interplanetary Society*, 33, 251–264.

Freitas, R. A., & Merkle, R. C. (2004). *Kinematic Self-Replicating Machines.* Landes Bioscience.

Frisch, P. C., Redfield, S., & Slavin, J. D. (2011). The interstellar medium surrounding the Sun. *Annual Review of Astronomy and Astrophysics*, 49, 237–279.

Gros, C. (2017). Universal scaling relation for magnetic sails: Momentum braking in the limit of dilute interstellar media. *Journal of Physics Communications*, 1, 045007.

Harris, T. E. (1963). *The Theory of Branching Processes.* Springer-Verlag (Grundlehren der mathematischen Wissenschaften, Band 119).

Hart, M. H. (1975). An explanation for the absence of extraterrestrials on Earth. *Quarterly Journal of the Royal Astronomical Society*, 16, 128–135.

Landis, G. A. (1998). The Fermi paradox: An approach based on percolation theory. *Journal of the British Interplanetary Society*, 51, 163–166.

Metzger, P. T., Muscatello, A., Mueller, R. P., & Mantovani, J. (2013). Affordable, rapid bootstrapping of the space industry and solar system civilization. *Journal of Aerospace Engineering*, 26(1), 18–29.

NASA (1982). *Advanced Automation for Space Missions* (R. A. Freitas & W. P. Gilbreath, Eds.; NASA Conference Publication 2255). Proceedings of the 1980 NASA/ASEE Summer Study.

Perakis, N., & Hein, A. M. (2016). Combining magnetic and electric sails for interstellar deceleration. *Acta Astronautica*, 128, 13–20.

Poston, D. I., Gibson, M. A., Sanchez, R., & McClure, P. R. (2020). Results of the KRUSTY nuclear system test. *Nuclear Technology*, 206(sup1), S89–S117.

Redfield, S., & Linsky, J. L. (2008). The structure of the local interstellar medium. IV. Dynamics, morphology, physical properties, and implications of cloud-cloud interactions. *The Astrophysical Journal*, 673, 283–314.

Tipler, F. J. (1980). Extraterrestrial intelligent beings do not exist. *Quarterly Journal of the Royal Astronomical Society*, 21, 267–281.

von Neumann, J., & Burks, A. W. (1966). *Theory of Self-Reproducing Automata*. University of Illinois Press.

This is the fourth document in the series and the first to be primarily quantitative. It supplies first-order budgets — mass and power, braking, reproduction demographics, and resource closure — that the conceptual papers deferred, and sets up the computational model (R_eff, failure and viability distributions, and closure across the real catalogue) that the companion computational paper carries out. The seed-to-factory capability ladder and the biological-capability question are treated separately in the bootstrapping paper and are deliberately not duplicated here.